Participant workbook

Introduction to AI-Assisted Research Methods and Data Analysis

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Welcome

Over four days you will take one clinical question from a hunch to an auditable mini research report. You do not need to know how to code. You will learn to read, run and adapt code, to use an AI assistant safely, and to check every number before you trust it.

How this workbook fits with the other materials

Material	Use it for
Slides	The ideas, demonstrated live
Notebooks (Python or R)	The hands-on analysis; exercises are marked inside
This workbook	Exercise instructions, written answers, checklists, reference pages
Word templates (templates/)	Worksheets and logs you fill in and hand in

Choose one language. Every notebook exists in Python and in R and gives the same results. If you have never coded, pick either; if you already use one, use that.

Ground rules

- **All course data are synthetic.** The patients do not exist and the associations are simulated. Never cite them as evidence.
- **Never** paste identifiable patient data, confidential manuscripts or restricted full text into a public AI tool. Follow your institution's policy.
- An AI-supplied reference is a **task to verify**, not evidence.
- AI code that runs is not the same as AI code that is right.
- You are responsible for every source, decision and number you report.

The course at a glance

Day	You will	You will leave with
1	Frame a question, choose a design, map variables, search and verify literature	Question, hypotheses, framework map, search log, literature matrix, annotated entry
2	Read data formats, use Copilot, clean data, handle missing data, describe variables	Clean data, cleaning log, analysis set, univariate pack

Day	You will	You will leave with
3	Estimate effects with CIs; compare groups; fit linear and logistic models	Bivariate results; adjusted estimates; interpretations
4	Make it reproducible; build tables, figure and report; write the discussion	Mini research report and handover package

Daily times: 08:30 to 16:00. Breaks 10:00 to 10:15, lunch 11:45 to 12:45, break 14:15 to 14:30.

Day 1: From clinical question to evidence map

Exercise 1A: Frame your clinical question (block 1)

Use `templates/Research_Question_Worksheet.docx`, part 1.

1. Write one clinical observation from your practice.
2. Turn it into PICO (an intervention you assign) or PECO (an exposure you observe).
3. Write H0 and H1. Check they name the same population, exposure, comparison and outcome as your question.
4. Do a FINER check. Revise the question once and note what changed.

Done when: one question plus hypotheses that refer to the same population and contrast.

AI assist (optional): *"I am a [role]. Here is my draft research question: [question]. Using PECO and FINER, list what is vague or missing. Do not rewrite it; ask me questions instead."*

Exercise 1B: Variable and framework map (block 2)

Use the worksheet, part 2.

1. Put the exposure and outcome on the page.
2. Add at least two confounders, one mediator and one possible collider.
3. Mark each construct as measured or unmeasured (dashed box).
4. Name one framework or theory to interpret your findings, and which of its constructs you can measure.
5. Write one limitation the map reveals.

Remember: adjust for confounders; do not adjust for mediators or colliders.

Search and Zotero practicum (block 3)

1. Build concept blocks: free-text synonyms and MeSH terms for each concept.
2. Combine with OR inside a block and AND between blocks.
3. Run the search in PubMed. Record it in `literature/search_log_template.csv` straight away.
4. Import three relevant records into a Zotero collection (browser button, or **Add Item by Identifier** with the DOI or PMID).
5. Check each record's metadata against the publisher page. Fix errors.
6. Open one PDF in Zotero, highlight and annotate, then **Add Note from Annotations**. Tag the note with a framework construct.

Done when: a search log entry, three verified records and linked PDF notes.

Exercise 1C: Literature mini-set (block 4)

Moving sources from Zotero to NotebookLM

1. In Zotero, select your verified items. Drag their **PDF attachments** to a folder on your computer.
2. Optionally right-click your annotation note and **Export Note** as Markdown.
3. In NotebookLM, create a notebook and **add sources**: upload the PDFs and the note.
4. Keep Zotero as the master record. Cite from Zotero, never from NotebookLM.

Then

1. Ask three **bounded** questions (for example *"For each source, how was adherence measured? Quote the passage."*).
2. For each answer, click the citation and run the five checks below.
3. Fill three rows of `literature/literature_matrix_template.csv`.
4. Write one annotated bibliography entry (`templates/Annotated_Bibliography_Template.docx`).
5. Log your AI use.

The five checks

Check	Question
Entity	Right paper, population and variable?
Passage	Does the cited passage actually say this?
Context	Main finding, subgroup, limitation, or someone else's result?
Numbers	Do estimates, CIs and sample sizes match the paper's table?
Record	Is the source in Zotero with verified metadata?

No access to NotebookLM? Use the printed excerpts and answer the same bounded questions by hand. The checks are the same.

Day 1 exit checklist

- ☐ Question with matched H0 and H1
- ☐ Variable and framework map
- ☐ Search log entry and three verified Zotero records with notes
- ☐ Three verified matrix rows and one annotated bibliography entry
- ☐ Notebook 00 runs and prints *Setup check passed* (do this tonight if needed)

Day 2: From raw export to analysis-ready data

Notebooks 00 and 01 (block 1)

1. **Notebook 00:** open the course folder in VS Code, open the notebook, select the kernel, **Run All**. Complete Exercise 0.
2. **Notebook 01, Part A:** read the Excel, SPSS, Stata and REDCap-style files; reshape long to wide. Complete Exercise 1A.
3. **Notebook 01, Part B:** use the five-part prompt in Copilot Chat. Complete Exercise 1B: audit the AI-generated average.
4. Add a row to your AI-use log.

Exercise 1B: would you report the AI's number? Write two or three sentences here or in the notebook.

Exercise 2A: Audit and defend a cleaning rule (block 2)

1. Run the audit cells. List the problems you find.
2. **Part 1:** standardise `smoking_status`; check exactly three categories remain.
3. Work through dates, ranges, duplicates and derived variables.
4. **Part 2:** choose one rule and write a four-sentence defence:

Sentence	Content
1	What you saw
2	The rule you applied
3	How many values or rows it affected
4	What the rule could get wrong

Done when: `data/processed/primary_care_clean.csv` is saved and every assertion passes.

Exercise 2B part 1: Who did we lose? (block 3)

1. Make the missingness table.
2. Look for clues: HbA1c by diabetes and clinic; adherence missing versus present.
3. Build the analysis set and flow table.
4. Compare included and excluded patients on age, diabetes and SBP.
5. Write one limitation sentence linking missing data to its likely consequence (selection, bias or precision).

Exercise 2B part 2: Univariate evidence pack (block 4)

One table and two figures describing BMI, clinic and uncontrolled BP. Then a five-sentence interpretation: give denominators, describe without explaining.

Day 2 exit checklist

- ☐ `primary_care_clean.csv` and the cleaning log
- ☐ Analysis-set flow and one missing-data limitation
- ☐ Univariate table, two figures and interpretation
- ☐ At least two AI-use log entries

Day 3: Estimation, comparison and regression

Exercise 3.0: Choose the method (block 1)

For each question, name the outcome type, exposure type, bivariate method and effect measure. Then complete `templates/Analysis_Plan_Template.docx` for your own question.

Question	Outcome type	Exposure type	Method	Effect measure
Mean BMI, current vs never smokers				
Uncontrolled BP across clinics				
Adherence and uncontrolled BP, adjusting for age and diabetes				

Exercise 3A: Defend a bivariate analysis (block 2)

Part 1: SBP by diabetes. Group sizes, means, boxplot; Welch test; difference (diabetes minus no diabetes) with 95% CI and p-value.

Part 2: uncontrolled BP by diabetes. Cross-table with row percentages; chi-square; difference in proportions in percentage points.

Interpret each in one or two sentences, with one limitation.

Exercise 3B: Fit, check and interpret a linear model (block 3)

1. Fit the crude and adjusted models.
2. Make a small table comparing the adherence estimate and CI in both.
3. Make the diagnostic plots and write a two-sentence diagnostic note.
4. Interpret the adjusted estimate in one sentence a colleague could read. Include *"among patients with the same…"*.

Exercise 3C: A defensible logistic interpretation (block 4)

1. Fit the adjusted logistic model.
2. Report odds ratios with 95% CIs.
3. Calculate the average predicted probability at adherence 40, 60 and 80.
4. Write three sentences: the adjusted OR with CI; what it means; what the predicted probabilities add.

Check your wording: odds, not risk; *associated with*, not *reduces*.

Day 3 exit checklist

- ☐ Analysis plan with estimand and effect measure
- ☐ Two bivariate comparisons with effect sizes and CIs
- ☐ Crude and adjusted linear estimates and diagnostic note
- ☐ Adjusted OR, predicted probabilities, three-sentence interpretation

Day 4: Reporting, discussion and capstone

Reproducibility audit (block 1)

For Notebooks 02 and 03: **Restart**, **Run All**, and note any error.

Question	Answer
Did both run without errors?	
Did any number change since yesterday?	
One result that depends on something outside the notebook, and how it is protected	

Exercise 4A and results pack (block 2)

Weak draft: *"Adherence was highly significant ($p<0.001$) for blood pressure, proving that taking medication lowers BP. Patients with good adherence had 19% less uncontrolled BP."*

What is wrong?

Your rewrite (two or three sentences, using Table 2):

Then build Table 1, Table 2 and the figure in Notebook 04, and render `reports/mini_report.qmd`.

Structured discussion (block 3)

Use `templates/Discussion_Matrix_Template.docx`.

Label	Part	Ask yourself
D	Direct answer	What did we find, with the main estimate?
L	Literature	How does it compare with my verified sources, and why might it differ?
F	Framework	What could it mean? What did I not measure?

Label	Part	Ask yourself
A	Alternatives and limitations	Confounding, selection, measurement, missing data: direction and likely size
R	Recommendation	What should practice or the next study do, given this strength of evidence?

Then run the claim and evidence audit: every factual sentence gets a location, or is softened or deleted.

Capstone (block 4)

Brief. In teams of two or three, using the synthetic dataset (or an approved equivalent), submit a two-page-equivalent mini research report with:

- question, hypotheses and framework map
- three verified literature-matrix entries and one annotated bibliography entry
- cleaning decisions and the analysis-set flow
- Table 1, one main results table and one figure
- a D/L/F/A/R discussion
- an AI-use and verification disclosure
- the notebook and all files needed to regenerate the outputs

Rubric

Criterion	Weight	Full-credit evidence
Question and framework	15	Focused question, matched hypotheses, coherent map, explicit inference boundary
Literature	15	Traceable search, verified sources, accurate source-to-claim links
Data and analysis	25	Documented cleaning, suitable models, diagnostics, missing-data decision
Results and discussion	25	Accurate estimates and CIs, clear table and figure, proportionate interpretation
Reproducibility and AI use	20	Run All succeeds, outputs regenerate, disclosure and verification record

90 to 100 exemplary and auditable · 75 to 89 secure · 60 to 74 developing · below 60 incomplete.

Handover checklist

- ☐ Restart and Run All succeeds
- ☐ Report renders; every number matches the notebook
- ☐ Every literature claim links to a verified source

- ☐ AI-use log complete
- ☐ No identifiable or confidential data anywhere

Reference: working with an AI coding assistant

GitHub Copilot in VS Code

Mode	How
Inline suggestion	Type a comment such as # count patients per clinic, pause, press **Tab** to accept
Copilot Chat	Open the chat panel; ask in plain language
Explain	Select code, right-click, **Copilot > Explain**

Copilot has a free plan; students and teachers can apply for more through GitHub Education (check current terms). If you already pay for another assistant (Gemini Code Assist, Claude, ChatGPT, Cursor), you can use it. Google Colab (browser) has Gemini built in.

The five-part prompt

Part	Include
Goal	The research task in one sentence
Data	Data frame name, variable names, types, units. **No real patient values.**
Constraints	Language and packages; short and commented
Output	What you want back (a table, a plot, a number)
Check	Ask how to confirm the result is right

Checking AI code

- **Denominator:** how many rows or values went in? Were any silently dropped?
- **Range:** do minimum and maximum make clinical sense?
- **Duplicates:** does anyone count twice?
- **Direction:** which group was subtracted from which? Which level is the reference?
- **Explain:** can you say what every line does? If not, ask Copilot to explain it.

Reference: code patterns

The notebooks contain complete code. These patterns help you adapt it to your own data.

Task	Python (pandas)	R (tidyverse)
Read a CSV	pd.read_csv(path)	read_csv(path)
Read Excel, skip 3 rows	pd.read_excel(path, skiprows=3)	readxl::read_excel(path, skip = 3)
First rows	df.head()	head(df)
Count a category	df["clinic"].value_counts()	count(df, clinic)
Filter rows	df[df["age"] >= 65]	filter(df, age >= 65)
New variable	df["adherence_10"] = df["medication_adherence"] / 10	mutate(df, adherence_10 = medication_adherence / 10)

Task	Python (pandas)	R (tidyverse)
Missing per column	df.isna().sum()	colSums(is.na(df))
Mean by group	df.groupby("diabetes")["systolic_bp"].mean()	df |> group_by(diabetes) |> summarise(mean(systolic_bp))
Welch t-test	stats.ttest_ind(a, b, equal_var=False)	t.test(a, b)
Chi-square	stats.chi2_contingency(pd.crosstab(x, y))	chisq.test(table(x, y))
Linear model	smf.ols("y ~ x + z", data=df).fit()	lm(y ~ x + z, data = df)
Logistic model	smf.logit("y ~ x + z", data=df).fit()	glm(y ~ x + z, data = df, family = binomial)
Odds ratios	np.exp(model.params)	broom::tidy(model, exponentiate = TRUE, conf.int = TRUE)
Stop if a rule breaks	assert df["patient_id"].is_unique	stopifnot(!anyDuplicated(df$patient_id))

Glossary

Term	Meaning
Analysis set	The patients and variables actually used in the models
Assertion	A line of code that stops the notebook if a rule is broken
Collider	A variable caused by both exposure and outcome; adjusting for it creates false associations
Complete-case analysis	Using only patients with no missing values in the variables needed
Confidence interval (95%)	Range of values for the true effect reasonably compatible with the data
Confounder	A variable that affects both exposure and outcome
Data dictionary	Description of each variable: meaning, type, valid values, notes
Estimand	The exact quantity a study sets out to estimate
Kernel	The engine that runs a notebook's code (Python or R)
MCAR / MAR / MNAR	Missing completely at random / at random given recorded data / not at random
Mediator	A variable on the causal path from exposure to outcome
Odds ratio	Ratio of the odds of the outcome in two groups; not the same as a risk ratio
p-value	How surprising the data would be if there were no true effect
PICO / PECO	Population, Intervention or Exposure, Comparison, Outcome

Term	Meaning
Quarto	Software that turns text, tables and figures into reports and slides
Reference level	The category others are compared with in a regression model
Reproducible	Someone else can regenerate every result from your files
Tidy data	One row per observation, one column per variable, one value per cell
Welch t-test	Compares two means without assuming equal variances